

Basics of Innate Immunity II

TLRs, inflammasomes, neutrophil recruitment/killing, phagocyte disorders, and IL-12/IFN-gamma

If you remember nothing else

- TLRs 2, 4, 3, 7, 8, and 9 are specifically emphasized as innate pattern-recognition receptors.
- Inflammasome activation links cytosolic danger sensing to inflammatory cytokine processing and autoinflammatory disease.
- Neutrophils must circulate, exit blood, ingest organisms, and kill them; defects can occur at different steps.
- The IL-12 / interferon-gamma axis is a major innate immune communication pathway.

Neutrophil trafficking

- High-yield sequence: selectin-mediated rolling -> integrin-mediated firm adhesion -> diapedesis -> chemotaxis.
- After tissue entry, phagocytosis and intracellular killing become the key tasks.
- Do not confuse a trafficking defect with a respiratory-burst or lysosomal-trafficking defect.

Phagocyte disorder anchors

- Chronic granulomatous disease: defective oxidative killing / respiratory burst.
- Chédiak-Higashi syndrome: lysosomal trafficking defect; the lecture highlights giant granules and pigmentation abnormalities as distinguishing clues.

Rapid recall

- TLR signaling defects can produce human inborn errors of immunity.
- Inflammasome questions usually test inflammatory signaling, not antibody function.
- IL-12 and IFN-gamma should be remembered as a linked macrophage/NK/T-cell communication axis.